NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
FIRST YEAR FALL SEMESTER (SOFTWARE ENGINEERING)
MID-TERM EXAMINATION 2022
BATCH 2021-2022

Total Marks: 20 **Code A** **Time: 90 mins**

MT-173 (Calculus)

Q1.	Determine whether $\lim_{x\to2} f(x)$ and $\lim_{x\to4} f(x)$ exist or not. (CLO-1) $f(x)=\begin{cases}2x+1 & if\ \ 0\le x\le 2\\ 7-x & if\ \ 2\le x\le 4\\ x & if\ \ 4\le x\le 6\end{cases}$ Also, sketch the graph.	(4)
Q2.	Check by the following general terms, whether the sequence is convergent or not ? (CLO-3) a) $T_n = \frac{(3n-1)(n^4-n)}{(n^2+2)(n^3+1)}$ b) $T_n = \left(1+\frac{4}{n}\right)^{4n}$	(4)
Q3.	(a) Express $z=-2-7i$ in polar form. Draw its Argand diagram along with its mirror image. (CLO-3)	(2)
	(b) Prove that (CLO-3) $cos^5\theta sin^7\theta = -\frac{1}{2^{11}}(sin12\theta - 2sin10\theta - 4sin8\theta + 10sin6\theta + 5sin4\theta - 20sin2\theta)$	(6)
Q4.	Find limits. (CLO-1) (a) $\lim_{x\to0}\left(\frac{1}{x\sqrt{1+x}} - \frac{1}{x}\right)$ (b) $\lim_{x\to0}\left(\frac{1-cos5x}{1-cos7x}\right)$	(4)

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.
Use pencil for any diagrams or graphs.
Questions are to be attempted on **blank A4 pages** only.

Answer **all** questions.
Essential working must be shown for full marks to be awarded.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets at the end of each question.
The total number of marks for this paper is 20.

FOR EXAMINER'S USE

15/20

Course Teacher : Faizan Ahmed Faiz

Question 4

Find limits

a) $\lim\limits_{x \to 0} \left(\frac{1}{x\sqrt{1+x}} - \frac{1}{x} \right)$

$\lim\limits_{x \to 0} \frac{x - x\sqrt{1+x}}{x^2\sqrt{1+x}}$

$= \frac{\cancel{x}(1-\sqrt{1+x})}{x^2(\sqrt{1+x})}$

$\frac{(1-\sqrt{1+x})}{x\sqrt{1+x}} \times \frac{1+\sqrt{1+x}}{1+\sqrt{1+x}}$

$\frac{1-(\sqrt{1+x})^2}{(x\sqrt{1+x})(1+\sqrt{1+x})}$

$\frac{\cancel{1}-\cancel{1}-x}{(x\sqrt{1+x})+x(\sqrt{1+x})^2}$

$\frac{-x}{x\sqrt{1+x}+x(1+x)}$

$\frac{-x}{x\sqrt{1+x}+x+x^2}$

$\frac{-\cancel{x}}{x(\sqrt{1+x}+1+x)}$

$= \frac{-1}{\sqrt{1+x}+1+x}$

2

A.L

$\frac{-1}{\sqrt{1+0}+1+0} = \frac{-1}{1+1} = \boxed{\frac{-1}{2}}$

PART (a):

$\frac{d}{dx} y^2 \cdot \frac{dy}{dx}$

...ER NO: 4 PART (b)

$$\left(\frac{1-\cos 5x}{1-\cos 7x}\right)$$

$$\lim_{x\to 0} \frac{\cancel{2}\sin^2\frac{5x}{2}}{\cancel{2}\sin^2\frac{7x}{2}}$$ $\therefore \sin^2\frac{x}{2} = \frac{1-\cos x}{2}$

$$= \lim_{x\to 0} \frac{\left(\sin\frac{5x}{2}\right)^2}{\left(\sin\frac{7x}{2}\right)^2}$$

$$= \lim_{x\to 0} \frac{\left[\frac{\sin\frac{5x}{2}}{\frac{5x}{2}} \times \frac{5x}{2}\right]^2}{\left[\frac{\sin\frac{7x}{2}}{\frac{7x}{2}} \times \frac{7x}{2}\right]^2}$$ $\therefore \frac{\sin\theta}{\theta} = 1$

$$= \lim_{x\to 0} \frac{1 \times \left(\frac{5x}{2}\right)^2}{1 \times \left(\frac{7x}{2}\right)^2}$$

$$\Rightarrow \cancel{\lim_{x\to 0}} \frac{\frac{5^2 \cancel{x^2}}{\cancel{2^2}}}{\frac{7^2\cancel{x^2}}{\cancel{2^2}}}$$

$= \cancel{25/49}.$

$$= \frac{5^2}{7^2} \lim_{x\to 0} \frac{\cancel{x^2}}{x^2} \times \frac{1}{1}$$

$$= \frac{25}{49}.$$

2

$$\lim_{x \to 4} f(x)$$

First

$$\lim_{x \to 4^-} (7-x) = 7-4 = \boxed{3}$$

Then

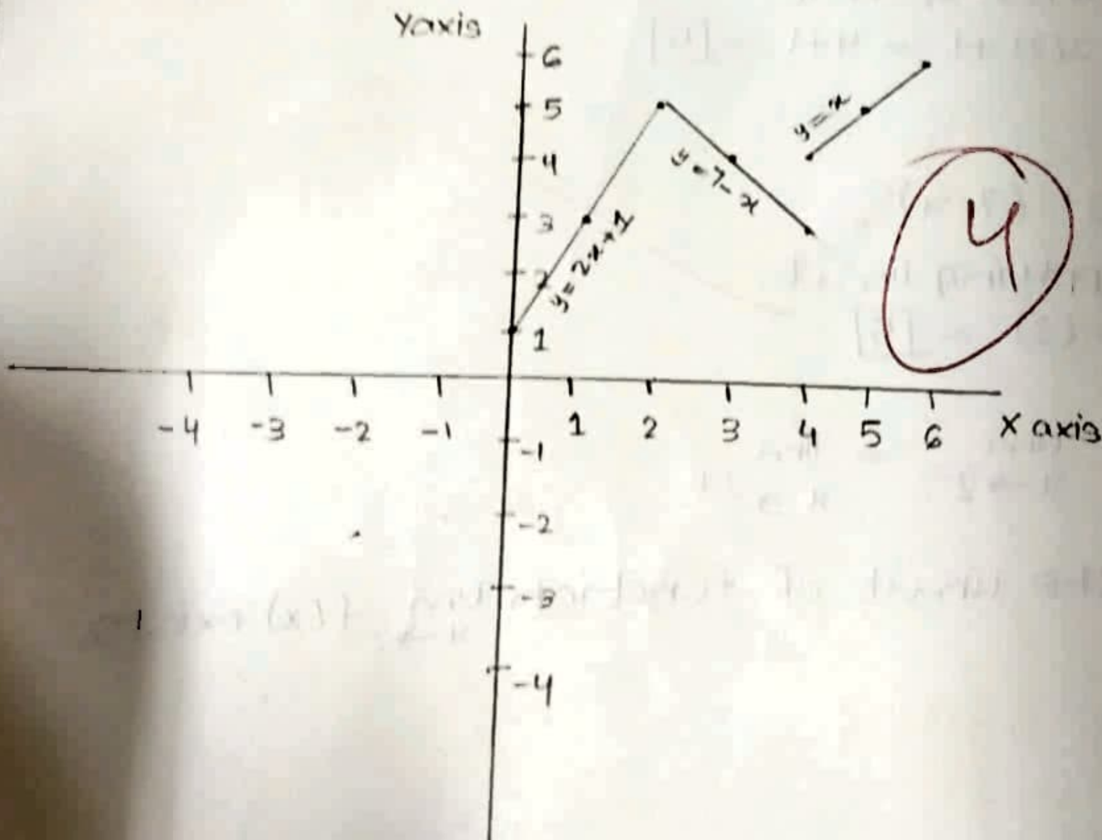
$$\lim_{x \to 4^+} x = 4 = \boxed{4}$$

Since $\lim_{x \to 4^-} \neq \lim_{x \to 4^+}$

Therefore limit does not exist.

GRAPH:

$$f(x) = \begin{cases} 2x+1 & \text{if } 0 \le x < 2 \\ 7-x & \text{if } 2 \le x \le 4 \\ x & \text{if } 4 \le x \le 6 \end{cases}$$



4

Identify the formula ~~of~~ for the function graphed.

Apply relevant

a) $\lim_{x \to 0} \frac{\ln \cos x}{\ln(\cos 3x)}$ b) $\lim_{x \to 0} x \tan\left(\frac{\pi}{2} - x\right)$

Q3 You have spherical storage tank containing oil
The tank has diameter of 6ft
You are asked to calculate the height ------

$$f(h) = h^3 - 9h^2 + 3.8197 = 0$$

(3 decimal places.)

Q4) Assuming the possibility of expansion find series in $x+1$ of

$$f(x) = e^{-x}$$

SEAT NO. SE-22023

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

FIRST YEAR(SOFTWARE ENGINEERING)
FALL SEMESTER EXAMINATIONS 2022
BATCH 2022

Time: 3 Hours **Dated:07-02-2023**
Max.Marks:60

Calculus - MT-173

Instructions:

1. Candidates are required to attempt **ALL** questions.
2. All questions carry equal marks.
3. Candidates should show all working in the answer sheet provided with the question paper. Omission of essential working will result in loss of marks.

CLO Distribution	Total Marks
Q.1,2,3 (CLO2)	36
Q.4,5 (CLO1)	24

Q1. **(a)** Apply reduction formula to evaluate $\int cot^6 x\, dx$ (4)

(b) Prove that:

$$\int_0^{\frac{\pi}{2}} sin^p x cos^q x dx = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}$$ (4)

(c) Evaluate $\int_0^{\infty} x^{\frac{1}{6}} e^{-\sqrt{\frac{x}{2}}} dx$ (4)

Q2. **(a)** Apply L'Hopital to evaluate the following indeterminate forms.

i) $\lim\limits_{x\to\frac{\pi}{4}}(1-tanx)sec2x$ ii) $\lim\limits_{x\to\frac{\pi}{2}}(1-sinx)^{cosx}$ (4)

(b) Calculate the 5th root of 17 by using Newton-Raphson Method (upto three decimal places). (4)

(c) If $x = a(\emptyset + sin\emptyset)$ *and* $y = a(1-cos\emptyset)$, then show that the radius of curvature is $\rho = 4acos\frac{\emptyset}{2}$. (4)

Q3. **(a)** If $z(x,y) = 7x^2y^4 - 2xy^2 + 4x^2$, find the percentage error in z if x changes from 3 to 2.93 and y changes from 4 to 4.45. (4)

(b) Apply leibnitz theorem on x^5e^{2x} to find the 5th derivative. (4)

(c) If $a = 7i - 3j + k,\ \ b = 9i - 2j - 3k,\ \ c = -4i - 6k$.
Show that $(a \times b) \times c \neq a \times (b \times c)$ (4)

Q4. **(a)** Solve the equation $x^6 + 1 = 0$, by using De-moivre's theorem. (4)

(b) If $z = aCos\theta + iaSin\theta$, prove that $\left(\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right) = 2\cos 2\theta$ (4)

(c) Express $cos^5\theta sin^7\theta$ in series of sines of multiples of θ (4)

Q5. **(a)** Sketch the graph of the following function:

$$f(x) = \begin{cases} -\frac{1}{2}(x+4)^2 + 3; & x \le -1 \\ 3|x-2| - 4; & -1 < x < 4 \end{cases}$$ (4)

(b) Test the convergence of the following: (4)

$$U_n = \frac{(3n-1)(n^4-n)}{(n^2+n)(n^3+1)}$$

(c) Discuss the continuity of the given function $f(x)$ at $x = 3$ (4)

$$f(x) = \begin{cases} x^2 + 3x & , \quad x \le 0 \\ x & , \quad 0 < x \le 3 \\ 3x^2 & , \quad x > 3 \end{cases}$$

GOOD LUCK !

...EERING & TECHNOLOGY

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

FIRST YEAR FALL SEMESTER (SOFTWARE ENGINEERING/ BACHELOR OF SCIENCE IN INDUSTRIAL CHEMISTRY) EXAMINATIONS 2018

BATCH 2018

Time: 3 Hours

Dated: 11-02-2019

Max.Marks: 60

Calculus - MT-173

Q#1: a) Solve the following (CLO-2)

i) Write the following expression in polar and exponential form: (4)

$\left(\frac{1-\sqrt{3}i}{1+\sqrt{3}i}\right)^6$

ii) If $\log\sin(x+iy) = u+iv$, prove that $\cosh 2y = \cos 2x + 2e^{2u}$ OR

$\sinh(z_1 - z_2) = \sinh z_1 \cosh z_2 - \cosh z_1 \sinh z_2$

b) Apply De Moivre's theorem to show that $\cos 5x = \cos x(16\cos^4 x - 20\cos^2 x + 5)$ (CLO-3) (4)

c) Solve $\tan^{-1} z = \frac{1}{2i}\log\left(\frac{1+iz}{1-iz}\right)$ (CLO-2) (4)

Q#2: a) Find Taylor series of $\sqrt{x}$ centered at x=1. (CLO-1) (4)

b) Discuss Continuity at x=1 and x=-1 of $f(x) = \begin{cases} 2x+1 & -3 \le x < -1 \\ x+2 & -1 \le x < 1 \\ 3x & 1 \le x < 3 \end{cases}$ (CLO-2) (4)

c) Apply Leibnitz theorem to Differentiate n times $(1+x^2)y'' + 2xy' - 5y = 0$, show that

$(1+x^2)y^{n+2} + 2(n+1)xy^{n+1} + (n^2+n-5)y^n = 0$ (CLO-3) (4)

Q#3: a) Apply L Hopital to find the limit of $\lim_{x\to 0}\left(\frac{1}{\sin x} - \frac{1}{x}\right)$. (CLO-3) (2)

b) Find the radius of curvature of $y = 4\sin x - \sin 2x$ at $x = \pi/2$. (CLO-1) (4)

c) Find the relative extrema of $f(x) = x^4 + 6x^2 - 2$ using second derivative test. (CLO-1) (3)

d) Find the relative extrema of $f(x,y) = x^3 - y^2 - xy + 1$ (CLO-1) (3)

Q#4: a) Apply reduction formula to evaluate $\int \tan^5 x\, dx$ (CLO-3) (4)

$\int \tan^n u\, du = \frac{1}{n-1}\tan^{n-1}u - \int \tan^{n-2}u\, du$

b) Find the convergence of the given improper integral $y = \frac{1}{x}$ from $x = 1$ to $x = \infty$. (CLO-1) (4)

c) Solve $\int_0^\infty \sqrt{x}\, e^{-\sqrt[3]{x}} dx$ using Gamma integral. (CLO-2) (4)

Q#5: a) A scaler field $V = x+y+z$ exists over the surface S defined by $2x+2y+z=2$ bounded by $x=0, y=0, z=0$ in the first octant. Find the surface integral $\int V\, ds$ over this surface. (CLO-1) (6)

b) Solve triple integral $I = \iiint 12xy^2z^3 dv$ over the rectangular box G defined by the (CLO-2) inequalities $-1 \le x \le 2, 0 \le y \le 3, 0 \le z \le 2$. (6)

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NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
FIRST YEAR FALL SEMESTER (SOFTWARE ENGINEERING)
MID-TERM EXAMINATION 2022
BATCH 2021-2022

Total Marks: 20 **Code A** **Time: 90 mins** Mom Rauf (21)

MT-173 (Calculus)

Q1.	Determine whether $\lim_{x\to 2} f(x)$ and $\lim_{x\to 4} f(x)$ exist or not. (CLO-1) $f(x) = \begin{cases} 2x+1 & if\ 0 \le x \le 2 \\ 7-x & if\ 2 \le x \le 4 \\ x & if\ 4 \le x \le 6 \end{cases}$ Also, sketch the graph.	(4)
Q2.	Check by the following general terms, whether the sequence is convergent or not ? a) $T_n = \frac{(3n-1)(n^4-n)}{(n^2+2)(n^3+1)}$ b) $T_n = \left(1+\frac{4}{n}\right)^{4n}$ (CLO-3)	(4)
Q3.	(a) Express $z = -2-7i$ in polar form. Draw its Argand diagram along with its mirror image. (CLO-3)	(2)
	(b) Prove that (CLO-3) $\cos^5\theta\sin^7\theta = -\frac{1}{2^{11}}(\sin 12\theta - 2\sin 10\theta - 4\sin 8\theta + 10\sin 6\theta + 5\sin 4\theta - 20\sin 2\theta)$	(6)
Q4.	Find limits. (CLO-1) (a) $\lim_{x\to 0}\left(\frac{1}{x\sqrt{1+x}} - \frac{1}{x}\right)$ (b) $\lim_{x\to 0}\left(\frac{1-\cos 5x}{1-\cos 7x}\right)$	(4)

SEAT NO. SE-22023

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

FIRST YEAR(SOFTWARE ENGINEERING)
FALL SEMESTER EXAMINATIONS 2022
BATCH 2022

Time: 3 Hours **Dated:07-02-2023**
Max.Marks:60

Calculus - MT-173

Instructions:

1. Candidates are required to attempt ALL questions.
2. All questions carry equal marks.
3. Candidates should show all working in the answer sheet provided with the question paper. Omission of essential working will result in loss of marks.

CLO Distribution	Total Marks
Q.1,2,3 (CLO2)	36
Q.4,5 (CLO1)	24

Q1. (a) Apply reduction formula to evaluate $\int \cot^6 x\, dx$ (4)

(b) Prove that:

$$\int_0^{\frac{\pi}{2}} \sin^p x \cos^q x\, dx = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\,\Gamma\left(\frac{p+q+2}{2}\right)}$$ (4)

(c) Evaluate $\int_0^\infty x^{\frac{1}{4}} e^{-\sqrt{x}} dx$ (4)

Q2. (a) Apply L'Hopital to evaluate the following indeterminate forms.

i) $\lim_{x\to\frac{\pi}{4}}(1-\tan x)\sec 2x$ ii) $\lim_{x\to\frac{\pi}{2}}(1-\sin x)^{\cos x}$ (4)

(b) Calculate the 5^{th} root of 17 by using Newton-Raphson Metho[illegible]to three decimal places). (4)

(c) If $x = a(\emptyset + \sin\emptyset)$ and $y = a(1-\cos\emptyset)$, then show tha[illegible] d[illegible] of curvature is $\rho = 4a\cos\frac{\theta}{2}$. (4)

Q3. (a) If $z(x,y) = 7x^2y^4 - 2xy^2 + 4x^2$, find the percentage error in [illegible] f x changes from 3 to 2.93 and y changes from 4 to 4.45. (4)

(b) Apply leibnitz theorem on x^5e^{2x} to find the 5^{th} derivative. (4)

(c) If $a = 7i - 3j + k$, $b = 9i - 2j - 3k$, $c = -4i - 6k$. Show that $(a\times b)\times c \neq a\times(b\times c)$ (4)

Q4. (a) Solve the equation $x^6 + 1 = 0$, by using De-moivre's theorem. (4)

(b) If $z = a\cos\theta + ia\sin\theta$, prove that $\left(\frac{z}{\bar{z}}+\frac{\bar{z}}{z}\right) = 2\cos 2\theta$ (4)

(c) Express $\cos^5\theta\sin^7\theta$ in series of sines of multiples of θ (4)

Q5. (a) Sketch the graph of the following function:

$$f(x) = \begin{cases} -\frac{1}{2}(x+4)^2 + 3; & x \leq [illegible] \\ 3|x-2| - 4; & -1 < [illegible] 4 \end{cases}$$ (4)

(b) Test the convergence of the following: (4)

$$U_n = \frac{(3n-1)(n^4-n)}{(n^2+n)(n^3+1)}$$

(c) Discuss the continuity of the given function $f(x)$ at $x = 3$ (4)

$$f(x) = \begin{cases} x^2 + 3x & , & x \leq [illegible] \\ x & , & 0 < x [illegible] 3 \\ 3x^2 & , & x > [illegible] \end{cases}$$

GOOD LUCK !

SEAT NO. 21-055

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

FIRST YEAR(SOFTWARE ENGINEERING)
FALL SEMESTER EXAMINATIONS 2021
BATCH 2021

Time: 3 Hours | **Dated:12-03-2022**
Max.Marks:60

Calculus - MT-173

Instructions:

1. Candidates are required to attempt ALL questions.
2. All questions carry equal marks.
3. Candidates should show all working in the answer sheet provided with the question paper. Omission of essential working will result in loss of marks.

CLO Distribution	Total Marks
Q.1,2,3 (CLO2)	36
Q.4 (CLO3)	12
Q.5 (CLO1)	12

Q1. **(a)** Apply reduction formula to evaluate $\int \sin^8 x\cos^7 x\,dx$ **(4)**

(b) Prove that:

$$\int_0^{\frac{\pi}{2}} \sin^p x\cos^q x\,dx = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\,\Gamma\left(\frac{p+q+2}{2}\right)}$$ **(4)**

(c) Evaluate $\int_0^{\infty} x^4 e^{-x^2}\,dx$ **(4)**

Q2. **(a)** Apply L'Hopital to evaluate the following indeterminate forms:

i) $\lim\limits_{x\to\frac{\pi}{4}}(1-\tan x)\sec 2x$ ii) $\lim\limits_{x\to 0}(1+\sin x)^{\frac{1}{x}}$ **(4)**

(b) Calculate the cube root of 13 by using Newton-Raphson Method (upto three decimal places). **(4)**

(c) Show that at any point on the curve $r = ae^{\theta\cot\alpha}$, radius of curvature is $\rho = r\,\mathrm{cosec}\,\alpha$. **(4)**

Q3. **(a)** If $z(x,y) = 10x^2y^4 - 6xy^2 + 10x^2$, find the relative error in z if x changes from 3 to 2.95 and y changes from 4 to 4.25. **(4)**

(b) Apply leibnitz theorem on x^6e^{3x} **(4)**

(c) Express the Taylor's series of $\ln(x)\,.\,Sin(x)$ in $(x-2)$. **(4)**

Q4. **(a)** Solve the equation $x^7 - 1 = 0$, by using De-moivre's theorem. **(4)**

(b) If $a = 3i - j + 2k$, $b = 2i - 2j - k$, $c = 3i + j$. **(4)**

Show that $(a \times b) \times c \neq a \times (b \times c)$

(c) Prove that: $\cos^5\theta\sin^3\theta = -\frac{1}{2^7}(\sin 8\theta + 2\sin 6\theta - 2\sin 4\theta - 6\sin 2\theta)$ **(4)**

Q5. **(a)** Sketch the graph of the following function:

$$f(x) = \begin{cases} -2x-4 & ;\ x < -2 \\ x^2-2 & ;\ -2 \le x < 1 \\ 2 & ;\ x \ge 1 \end{cases}$$ **(4)**

(b) Test the convergence of the following: **(4)**

$$U_n = \frac{(3n-1)(n^4-n)}{(n^2+n)(n^3+1)}$$

(c) Discuss the continuity of the given function $f(x)$ at $x = 2$ *and* $x = 4$. **(4)**

$$f(x) = \begin{cases} -1+10x & :\ 0 \le x \le 2 \\ x+4 & :\ 2 \le x \le 4 \\ 2x & :\ 4 \le x \le 6 \end{cases}$$

GOOD LUCK !

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
FIRST YEAR FALL SEMESTER (SOFTWARE ENGINEERING)
MID-TERM EXAMINATION 2022
BATCH 2022-2023

Total Marks: 20

Time: 90 mins
Date: 8th Dec'2022

MT-173 (Calculus)

Q1.	(a) Express $z = \frac{8+i\sqrt{2}}{2-i\sqrt{2}}$ in polar form. Draw its Argand diagram along with its mirror image. (CLO-1)	(2)
	(b) Expand $sin^9\theta$ in series of sines of multiples of θ (CLO-1)	(4)
Q2.	Determine whether the limit of the given function $f(x)$ exist at x=2 ? (CLO-1) $$f(x) = \begin{cases} -2x^2; & x \geq 2 \\ -2x+2 & ; x < 2 \end{cases}$$ Also, sketch the graph.	(4)
Q3.	Check whether the following general terms form a convergent or a divergent sequence.	
	(a) $U_n = \frac{n^2}{n!}$ (CLO-1)	(2)
	(b) $U_n = -450^n$ (CLO-1)	(2)
Q4.	If $x^3 + y^3 = 3axy$; prove that $\frac{d^2y}{dx^2} = \frac{-2a^3xy}{(y^2-ax)^3}$. (CLO-2)	(6)